

Kepler-17b

R-2428

Benchmark run · workflow W8-benchmark-deep · record deep-bench_KEPLER-17_180627 · created 2026-08-22T00:20:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458296.75646 +/- 0.00082 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.04
Transit depth (Rp/Rs)^2	2e-07 +/- 0.00036 [%]
Orbital Inclination (inc)	89.74 +/- 1.71
Airmass coefficient 1 (a1)	1.205 +/- 0.086
Airmass coefficient 2 (a2)	-0.143 +/- 0.057
Scatter in the residuals of the lightcurve fit is	7.11 %
Best Comparison Star	#2 - [365, 289]
Optimal Aperture	2.4
Optimal Annulus	7.27
Transit Duration (day)	0.0859 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.04 vs 0.13031 (deviation 2.66σ)
Duration fit vs published	0.0859 d vs 0.0954 d (deviation 3.24σ)
Mid-time O–C	-1.6 min
Depth SNR	0.0
Residual scatter	7.11 %

Light curve

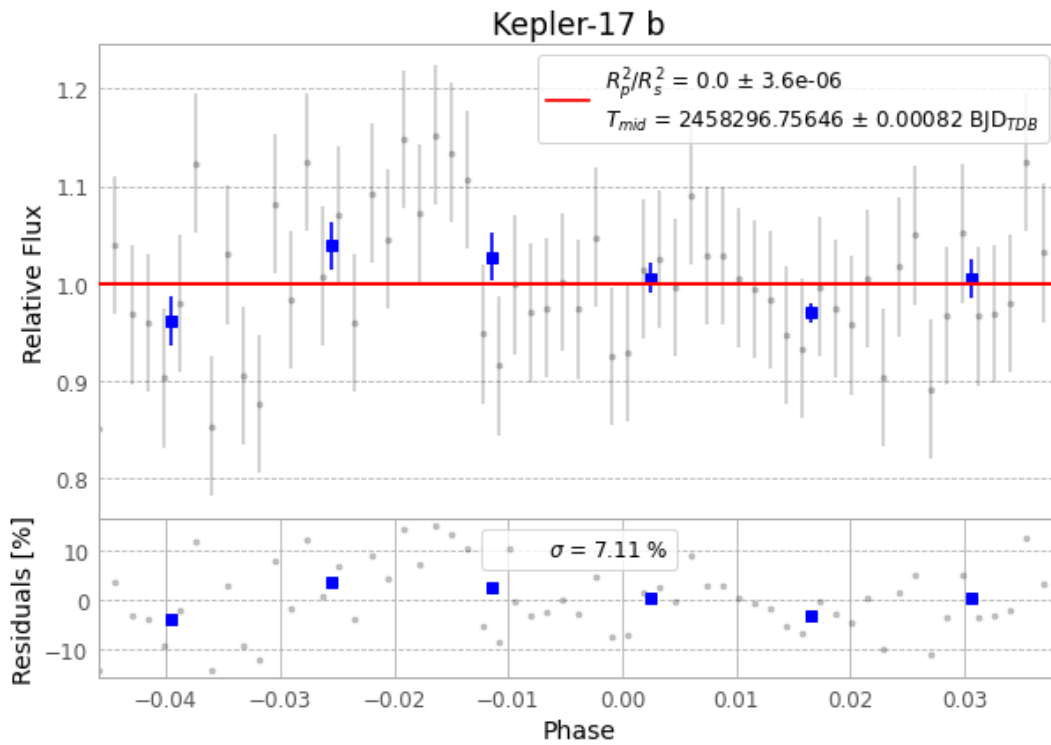


Figure 1 — FinalLightCurve_Kepler-17 b_2018-06-26.png (SHA-256 6ccdb9bdc8cc35ae...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 218], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2a36199bccaad7af...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), KEPLER-17180627042712.FITS → KEPLER-17180627072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KEPLER-17-180627.log (9c1d1627be84...), FinalLightCurve_Kepler-17 b_2018-06-26.png (6ccdb9bdc8cc...), FinalLightCurve_Kepler-17 b_2018-06-26.csv (ef9f61d1fe93...)

Compute resources

Wall time 102.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 00:20Z.